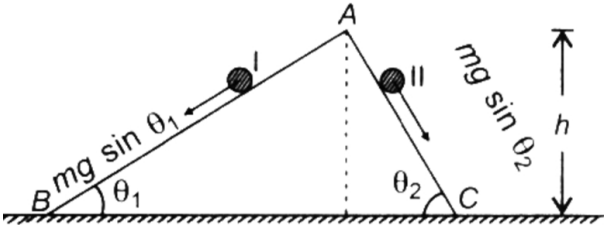


Solution

SOUND-06

Class 09 - Science

Section A

1. **(d)** erg
Explanation: The unit of work in the cgs system is dyne cm or erg.
2. **(a)** the momentum of the system is constant
Explanation: The total momentum of the system just before the collision is the same as the total momentum just after the collision. So, the momentum of the system is constant.
3. **(d)** remains the same
Explanation: The kinetic energy changes into the potential energy when stone rises up. The potential energy then changes back to the kinetic energy, when a stone falls down. A boy throws a ball vertically upwards. As the ball rises its total energy remains the same.
4. **(c)** a force of 1 N displaces a body by 1 m
Explanation: 1 joule(J) is the amount of work done when a force of 1 Newton(N) displaces an object 1 meter(m).
 $1J = 1N \times 1m$ or $1J = 1Nm$.
5. **(c)** $mg \frac{h}{2}$
Explanation:
 If you consider gravitational field to be constant till height 'h' then we know that Center of the mass of tank lies $\frac{h}{2}$ height above ground level.
 Now potential energy will be $= m \times g \times (\frac{h}{2})$.
6. **(b)** Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.
Explanation: AB and AC are two smooth planes inclined to the horizontal at different angles.
 As height of both the planes are same, therefore, both the stones will reach the bottom with same speed.
 $\therefore v = \sqrt{2gh} \therefore v_1 = v_2$

 From figure, acceleration, $a_1 = g \sin \theta_1$
 acceleration, $a_2 = g \sin \theta_2$
 As $\theta_2 > \theta_1$, so $a_2 > a_1$
 Using, $v = u + at$
 $v_1 = u_1 + a_1 t_1$; $v_1 = a_1 t_1$
 $t_1 = \frac{v_1}{a_1}$
 Similarly, $t_2 = \frac{v_2}{a_2}$ $\therefore a_2 > a_1$ So, $t_2 < t_1$
 Hence, stone II takes less time to reach the bottom.
7. **(a)** J/s
Explanation: Power is measured in energy (joules) divided by time. The SI unit of power is the watt (W) or joule per second (J/s). Power is a scalar quantity, it has no direction.

8. **(a)** potential energy
Explanation: Water stored in a dam possesses potential energy. And once the water is falling it has kinetic energy.
9. **(b)** Potential energy
Explanation: The energy possessed by an object because of its position is called potential energy. For example; when a stone is kept at a height, it possesses some energy because of its height. Because of this potential energy, the object kept at a height falls over the ground.
10. **(a)** 400 J
Explanation: Gravitational potential energy
 $E = mgh = 5 \times 10 \times 8 = 400 \text{ J}$
11. **(b)** Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.
Explanation: Relation between momentum and KE (E_K) is

$$p = \sqrt{2mE_k}$$
If mass of body decreases, then KE should increase accordingly to maintain constant momentum.
Also, $E_K = \frac{1}{2}mv^2$ or $E_K \propto v^2$
12. **(a)** zero
Explanation: The object moves in a circular path. In this case, the centripetal force acts on the object at right angles to the direction of motion of an object. So, the work done by the object in a circular path is zero.
13. **(a)** 10 m
Explanation: P.E. = mgh
 $m = 50 \text{ kg}$ P.E. = 4900J $g = 9.8 \text{ m/s}^2$
 $4900 = 50 \times 9.8 \times h$
 $4900 = 490 \times h$
 $h = \frac{4900}{490} = 10 \text{ m}$
14. **(c)** Electrical signal/energy
Explanation: Microphones are loudspeakers in reverse, it converts sound (mechanical) energy to electrical energy.
15. **(d)** 0.102 m
Explanation: P.E. = mgh
 $m = 1 \text{ kg}$ P.E. = 1J $g = 9.8 \text{ m/s}^2$
 $1 = 1 \times 9.8 \times h$
 $h = \frac{1}{9.8} = 0.102 \text{ m}$
16. **(b)** there is no component of force, parallel to the direction of motion
Explanation: Work is done whenever a force or a component of a force results in a displacement. No component of the force is acting in the direction of motion.
17. **(a)** 245 J
Explanation: P.E. = mgh
 $m = 5 \text{ kg}$ $h = 5 \text{ m}$ $g = 9.8 \text{ m/s}^2$
P.E. = $5 \times 9.8 \times 5$
P.E. = 25×9.8
P.E. = 245 J
18. **(c)** K.E. of the heavier is more than that of the lighter.
Explanation: P.E. of object 1 = mgh
P.E. of object 2 = 2mgh
On reaching the ground the P.E. of both the objects gets completely transformed into K.E.
Therefore K.E. of object 2 > K.E. of object 1.

19. (c) the body moves along the direction of the applied force
Explanation: The work done is positive if the applied force is in the same direction as the direction of motion.
20. (c) initial velocity of the object
Explanation: We know that, $W = F \cdot d \cos \theta$
 Here, F = force applied on the object, d = displacement and θ is angle between force and displacement. So, the work done on an object does not depend upon the initial velocity of the object.
- Section B**
21. (c) component
Explanation: Work can be either positive or negative: if the force has a component in the same direction as the displacement of the object, the force is doing positive work. If the force has a component in the direction opposite to the displacement, the force does negative work.
22. (b) mass, velocity
Explanation: Kinetic energy is the energy possessed by an object in motion. Kinetic energy is directly proportional to the mass of the object and to the square of its velocity: $K.E. = \frac{1}{2} m v^2$.
23. (d) increases, decreases
Explanation: When a body is thrown vertically upwards, during its upward movement the potential energy is increase due to increase in height and kinetic energy decreases due to decrease in velocity.
24. (b) velocity
Explanation: If the body is moving with instantaneous velocity V then instantaneous power is given by the relation: $P = F \cdot V$, this power is the dot product of force and velocity. It is a scalar quantity.
25. (a) buoyancy, the immersed
Explanation: Archimedes' principle states that the upward buoyant force that is exerted on a body immersed in a fluid, whether fully or partially submerged, is equal to the weight of the fluid that the body displaces and acts in the upward direction at the center of mass of the displaced fluid.
26. (b) increases
Explanation: $P \propto \frac{1}{A}$
27. (a) dives, floating tube
Explanation: In order to dive, the fish squeezes out air from its floating tube. This in turn increases its density and hence, it settles at the base of the river-bed, etc.
28. (c) capacity
Explanation: Energy is the capacity of doing work. An object which can do more work is said to have more energy and vice versa.
29. (d) $\frac{1}{3600}$
Explanation:
 1 Watt = 1 Joule Of Energy Consumed In 1 Second.
 i.e, $1 W = 1 J/S$
 Power consumed by the 60 watt bulb = 60 W
 $1 \text{ second} = \frac{1}{3600} \text{ hours}$
30. (d) increases

Explanation: When we compress the string potential energy increases.

$$U = \frac{1}{2} kx^2$$

x = compression, as x increase, U increase.

31. (a) less

Explanation: The weight of the sinker and cork combined, in water will be less than the weight of the sinker alone in water because the upthrust due to water on cork (when completely immersed) is more than the weight of cork itself.

32. (a) perpendicular

Explanation: Work is done only when a certain force is applied on an object and the object moves a certain distance on the direction of the applied force. Hence work done is zero when the direction is perpendicular.

33.

(c) component

Explanation: Work can be either positive or negative: if the force has a component in the same direction as the displacement of the object, the force is doing positive work. If the force has a component in the direction opposite to the displacement, the force does negative work.

34.

(d) 1 joule

Explanation: 1 Watt = 1 Joule per second and a 1 Joule = 1 Watt times one second.

35. (a) fluid, upthrust

Explanation: A fluid will exert a force upward on a body if it is partly or wholly submerged within it. This is because the deeper into a fluid you go, the greater the weight of it and so the greater the pressure. This difference in pressure between the top and the bottom of the object produces an upward force on it. This is called Upthrust.

36. (a) displaces

Explanation: When force is exerted on an object and object is displaced, work is said to be done.

37.

(d) Mean

Explanation: Potential energy is maximum at the extremes. When it again starts its motion towards the mean position then the potential energy is converted into kinetic energy. Thus at the mean position, the kinetic energy is maximum but potential energy is minimum. At the extreme position, the P.E is maximum but the K.E is minimum.

38.

(b) $\frac{11}{12}$, water

Explanation: It is found that an iceberg floats in seawater with $\frac{11}{12}$ part of its volume below water and $\frac{1}{12}$ part of its volume above seawater.

Section C

39.

(c) C and D

Explanation: (a) A bird sitting on tree possess potential energy only is correct statement.

(b) A stationary object may have energy is correct statement.

(c) A flying bird has kinetic energy only this is incorrect statement because A flying bird has kinetic energy as well as positional energy also.

(d) An aeroplane running on the run- way possess kinetic energy only. So given statement is incorrect.

40.

(d) A

Explanation: When an object starts to fall freely, the K.E. increase with a decrease in potential energy, and total mechanical energy remains constant. So the given statement is incorrect.

41. (a) A, C and D

Explanation: When an object starts to fall freely, the K.E. increase with decrease in potential energy and total mechanical energy remains constant. So, statement A, C, and D are incorrect and B is correct.

42.

(d) (A), (C) and (D) are incorrect

Explanation: When an object starts to fall freely, the K.E. increase with a decrease in potential energy, and total mechanical energy remains constant. So, statements A, C, and D are incorrect and B is correct.

43.

(c) C and D

Explanation:

A. A bird sitting on a tree possesses potential energy only is a correct statement.

B. A stationary object may have energy is a correct statement.

C. A flying bird has kinetic energy only this is the incorrect statement because A flying bird has kinetic energy as well as positional energy also.

D. An aeroplane running on the run-way possesses kinetic energy only. So the given statement is incorrect.

Section D

44.

(c) $P_1 = P_2$

Explanation: $P_1 = P_2$ as dimensions and material of cuboid is the same. Force exerted by each cuboid having the same area of contact is the same.

45. (a) Only student A

Explanation: $p \propto \frac{1}{A}$ (for the same magnitude of force)

If area increase then pressures decrease and if area decreases then pressure increase.

46.

(d) pressure increases with increases in force

Explanation: For same contact area, $P \propto F$

47.

(c) Student A only

Explanation: In both cases, cuboid will exert the same pressure but observation will be made by student A only.

48.

(d) same pressure in both cases

Explanation: The weight of the cuboid and base area is the same in both cases.

49. (a) the same force acts on an area equal to the contact area of a cuboid

Explanation: Force acting normally on a surface is called the thrust. The force acting normally (perpendicularly) on the unit area of a surface is called pressure. If F is the force acting normally on a surface of an area of cross-section A , then $P = \frac{F}{A}$.

50.

(c) equal to zero

Explanation: If $g = 0$ then $F = mg = 0$. Therefore, $p = 0$

51.

(c) upright, widest base

Explanation: In an upright position, the base area is smallest while in the widest base position, base area is greatest and $P \propto \frac{1}{A}$.

52.

(d) towards the pointed end of the pencil

Explanation: $P \propto \frac{1}{A}$

Smaller the area, greater is the pressure.

53. (a) the same conclusion is drawn by them

Explanation: Opposite parallel faces of cuboids have the same surface area of contact.

54.

(c) acceleration

Explanation: When both spheres are dropped simultaneously from a tower, they have same acceleration. Because, during free fall acceleration of body becomes equals to ($g = 9.8 \text{ m/s}^2$) and 'g' depends on mass of earth and radius of earth..

Section E

55. Three examples when force being supplied and still there is no displacement are:

- i. A man pushing a stationary truck.
 - ii. A man pushing a rigid wall.
 - iii. A boy carrying a basket on his head and standing still. Here, force of gravity acts on the basket, but there is no displacement.
- Three examples when force being supplied and still there is no displacement are: A man pushing a stationary truck. A man pushing a rigid wall. A boy carrying a basket on his head and standing still. Here, force of gravity acts on the basket, but there is no displacement.

56. Let, Potential energy of a person on Earth = mg_1h_1

Potential energy of same person on planet 'A' = mg_2h_2

Since, The potential energy of the person will remain the same on the Earth and on planet A.

Therefore, $mg_1h_1 = mg_2h_2$

Since mass remains same, So, if $g_1 = g$, then $g_2 = \frac{1}{2}g$

Here, $h_1 = 0.4$; Now $h_2 = \frac{g_1h_1}{g_2} = \frac{g \times 0.4}{\frac{g}{2}}$

or $h_2 = 0.4 \times 2 = 0.8$ m.

Therefore, he can jump on planet A = 0.8m

57. No, It doesn't violate the law of conservation of energy because the potential energy of a freely falling object decreases progressively since it keeps changing into kinetic energy of the free falling object therefore the total energy to the object remains conserved.

58. a. i. Potential energy

ii. Potential energy + Kinetic energy

iii. Kinetic energy

b. The initial energy is transformed into heat energy when the pendulum bob strikes the air molecules. Thus, the amplitude of pendulum decreases and finally it stops.

59. The man is doing work relative to the stream because he is applying force to produce relative motion between the boat and the stream. But he does no work relative to the shore as the displacement relative to the shore is zero.

60. Let, the mass of a ball = 'm'

The energy possessed by a ball at height (h) = $mgh = m \times 10 \times 10 = 100$ mJ. ($g = 10 \text{ ms}^{-2}$)

If energy is reduced by 40%, then the remaining energy = $100\text{mJ} - 40\text{mJ} = 60$ mJ.

Let h' be the height attained by a ball after bouncing back.

Now, Remaining energy = mgh'

Therefore, $60 \text{ m} = m \times 10 \times h'$ or $h' = 6$ m.

61. a) Work done is positive as the bucket moves in the direction of force applied by the man.

b) Work done by the gravitational force is negative, as the bucket moves upwards i.e. opposite to the gravitational force.

62. We have $P = 2 \text{ kW} = 2000 \text{ W}$, $g = 10 \text{ m/s}^2$, time = 2 minute = 60 s and height $h = 10$ m

We know; $P = \frac{W}{\Delta t}$ Or $P = \frac{mgh}{\Delta t}$

Or $2000\text{W} = \frac{m \times 10 \times 10}{60}$ Or $m = \frac{2000 \times 60}{10 \times 10} = 1200$ kg

So, the motor will pump 1200 kg water per minute.

63. i. Since in this diagram displacement is perpendicular to the direction of force, so work done is zero.

ii. Since in this diagram displacement is in the direction of force, so work done is positive.

iii. Since in this diagram displacement is in the opposite direction of the force applied hence work done is negative.

64. We know that, Power = $\frac{\text{Work done or energy}}{\text{time}} = \frac{mgh}{t} = m \cdot g \cdot \left(\frac{h}{t}\right)$

Since, speed = $\frac{\text{Distance}}{\text{time}} = \frac{h}{t}$

Therefore, mass, $m = \frac{\text{Power}}{g \times \text{Speed}}$

Hence, mass of a body that can be lifted = $\frac{\text{Power}}{g \times \text{Speed}} = \frac{100}{10 \times 1} = 10$ kg

65. Initial velocity of the car (u) = $60 \text{ km/h} = (60 \times 1000) / 60 \times 60 = 50/3 \text{ ms}^{-1}$

Final velocity (v) = 0 (object has to be stopped)

Initial kinetic energy = $\frac{1}{2} \times m \times v^2$

= $\frac{1}{2} \times 1500 \times (50/3)^2 = 208333.30 \text{ J}$

$$\text{Final kinetic energy} = \frac{1}{2} \times 1500 \times 0 = 0$$

$$\text{Therefore, work done} = \text{change in kinetic energy} = 208333.30 - 0 = 208333.30 \text{ J}$$

66. Given: Mass of water (m) = 5 kg

Height through which water is raised (h) = 10 m

Acceleration due to gravity (g): = 10 ms^{-2}

$$\therefore \text{Work done by the woman} = \text{P.E.} = mgh = 5 \times 10 \times 10 = 500 \text{ J}$$

$$\therefore \text{Power used by the woman} = P = \frac{W}{t} = \frac{500}{10} = 50 \text{ watt}$$

67. Initial velocity of stone (u) = 40 ms^{-1}

Final velocity of stone (v) = 0

Acceleration due to gravity (g) = -10 ms^{-2} [For upward direction g is -ve]

Height attained by stone (S) = ?

$$\text{We know, } v^2 - u^2 = 2gS$$

$$(0)^2 - (40)^2 = 2 \times (-10) \times s$$

$$\Rightarrow S = \frac{-1600}{-20} = 80 \text{ m}$$

$\therefore$ Maximum height attained by stone = 80 m

Net displacement of stone = 0

(since the stone returns back to the same point)

Total distance covered by the stone = $2 \times$ height attained

$$= 2 \times 80 = 160 \text{ m}$$

68. W.D. by the force when velocity is

$$5 \text{ ms}^{-1} = \frac{1}{2} mv^2 = \frac{1}{2} \times 20 \times (5)^2 = 250 \text{ J}$$

$$\text{W.D. by the force when velocity is } 2 \text{ ms}^{-1} = \frac{1}{2} mv^2 = \frac{1}{2} \times 20 \times (2)^2 = 40 \text{ J}$$

$$\text{Resultant W.D. by the force} = \text{Change in KE} = 40 - 250 = -210 \text{ J}$$

69. Commercial unit of energy = kilowatt hour i.e. kWh

SI unit energy = Joule (J)

$$\text{We know that; } 1 \text{ kWh} = 1 \text{ kW} \times 1 \text{ h} = 1000 \text{ W} \times 3600 \text{ s} = 1000 \text{ Js}^{-1} \times 3600 \text{ s} = 3600000 \text{ J} = 3.6 \times 10^6 \text{ J}$$

$$\text{Therefore, } 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

70. When salt is mixed with water it increases the density of water and, hence, its density becomes greater than that of the density of egg which now floats in it.

71. In case of a body moving along a circular path, the force (centripetal) is always along the radius while displacement is tangential.

$$\text{Hence work done } W = Fs \cos 90^\circ = 0 \text{ as angle between } F \text{ and } s \text{ is } 90^\circ.$$

72. Height of each step = 20 cm = 0.20 m

$$\text{Height of 30 steps, } H = 30 \times 0.20 \text{ m} = 6.0 \text{ m}$$

Time, t = 15 seconds

Force exerted by man against gravity

$$F = \text{Weight of man} = mg$$

$$\text{work done by man} = \text{Force} \times \text{Height} = mg \times H$$

Power

$$= \frac{mgH}{t}$$

$$= \frac{60 \times 10 \times 6.0}{15}$$

$$= 240 \text{ joule.}$$

73. Given, Power of motor, P = 40 kW = $40 \times 10^3 = 40,000 \text{ W}$

$$\text{Here, Load to be lifted} = \text{Force applied (F)} = 20,000 \text{ N}$$

If v is the speed of load, then we know that, P = Fv

$$\Rightarrow \text{Speed, } v = \frac{P}{F} = \frac{40,000}{20,000} = 2 \text{ m/s}$$

74. Since Work (W) = Force(F) $\times$ displacement(d)

$$= 7 \text{ N} \times 8 \text{ m} = 56 \text{ Nm or Joule}$$

75. Foundation of a building is made wider so that it may not sink under the extremely high pressure of building. The broad foundation distributes the weight of the building to the ground.